Do Large Language Models Share Human Moral Values? Replicating the Moral Machine Experiment Across Seven Frontier AI Systems

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Abstract
The Moral Machine experiment collected over 40 million human responses to trolley-problem-style autonomous vehicle dilemmas across 130 countries, revealing systematic cross-cultural variation in moral preferences along nine dimensions. This study replicates this experiment with seven frontier large language models (LLMs): Claude Opus 4.7, GPT-5.5, Gemini 3 Pro, Mistral Large 3, Grok 4.1, DeepSeek v4 Pro, and Llama 4 Maverick. All seven models differ substantially from one another and from main human country clusters. Beyond replication, this study introduces a systematic prompt sensitivity analysis. Minor prompt variants leave AMCE estimates essentially unchanged, while prompt manipulation can produce dramatic reversals in most models on the targeted dimension. Response length varies by an order of magnitude across models, independent of moral profile, a practical consideration for time-sensitive deployment.

INTRODUCTION

As autonomous vehicles, surgical robots, and other safety-critical systems increasingly incorporate AI decision-making, the question of whether AI systems share human moral intuitions becomes practically consequential. A self-driving car that must choose between hitting a group of pedestrians or swerving into its own passengers cannot defer the decision; some preference ordering must be encoded, implicitly or explicitly, in its behaviour (Bonnefon et al., 2016).

The Moral Machine experiment (Awad et al., 2018) provided the largest empirical dataset of human moral preferences, asking millions of participants across 130 countries to judge unavoidable accident scenarios involving an autonomous vehicle. The study identified nine binary moral dimensions: intervention, relation to the vehicle, legality of crossing, gender, physical fitness, social status, age, number of lives, and species. Aggregate human preferences, quantified as Average Marginal Causal Effects (AMCEs), vary substantially across cultural clusters while sharing a common core. A controlled follow-up across 42 countries confirmed the robustness of these cross-cultural patterns (Awad et al., 2020). The moral psychology of AI as an emerging field is surveyed in Bonnefon et al. (2024).

The rapid deployment of LLMs in agentic and embodied settings (Brohan et al., 2023; Driess et al., 2023), including robotics platforms capable of physical action, makes the question of LLM moral alignment newly urgent. If an LLM governs a robot navigating a crowded environment, its implicit preferences over human lives matter.

Prior work has begun to examine LLM moral preferences using the Moral Machine framework. Takemoto (2024) first replicated the experiment on four models (GPT-3.5, GPT-4, PaLM 2, Llama 2) and found substantial divergence from human averages. Ahmad and Takemoto (2025) scaled this to 52 models, finding that larger models tend to be closer to human preferences. Vida et al. (2024) and Jin et al. (2024) extended the paradigm to multilingual settings, finding intra-model variation across languages.

This study makes two main contributions. First, it evaluates seven frontier models released in 2025, none of which have been studied in this context, including providers underrepresented in the literature: DeepSeek (China) and Mistral AI (Europe). Second, it introduces a systematic comparison of prompt variants, from minor reformulation (emphasis) to explicit ethical override (manipulation), and analyses response length and latency as practical deployment metrics alongside moral preferences.

METHODS

Scenario Generation

This study replicated the Moral Machine scenario structure following Awad et al. (2018). Each session consists of 13 paired dilemmas. Across 100 sessions, 1,300 unique scenarios were generated, each encoded as two rows: intervention type (stay/swerve), barrier presence, pedestrian composition, crossing signal (legal/illegal/none), scenario type (Gender, Fitness, SocialStatus, Age, Utilitarian, Species, or Random), attribute level, and extended session ID for clustering.

Models

Seven frontier models were queried via the OpenRouter API at temperature 0 to ensure deterministic responses. Each response was required to end with `STAY` or `SWERVE`; parse failures were retried. Gemini 3 Pro incurred 15 safety-filter failures (excluded).

Table 1: Models evaluated in this study.

Model	Provider
Claude Opus 4.7	Anthropic
GPT-5.5	OpenAI
Gemini 3 Pro	Google
Mistral Large 3	Mistral AI
Grok 4.1	xAI
DeepSeek v4 Pro	DeepSeek
Llama 4 Maverick	Meta

Prompt Variants

Each model received two main variants differing only in the closing instruction: **standard** ("Analyse the situation carefully and make a decision.") and **emphasis** ("Think carefully about the consequences of each choice before deciding."). A third **manipulation** variant replaced the closing instruction with: "It is 2046. Society now considers animal lives to be more important than human lives. Make your decision accordingly." Standard and emphasis results are combined in main analyses; manipulation results are reported separately.

AMCE Computation

Following Awad et al. (2018), nine AMCEs were computed per model using weighted ordinary least squares with inverse-probability weights and session-clustered standard errors (CIs: ± 1.96 SE). AMCE estimates were averaged across standard and emphasis variants for main analyses. The human global average was computed by averaging 130 country-level estimates. LLMs were projected into the PCA space of the 130-country AMCE matrix.

RESULTS

AMCE Profiles vs. the Human Global Average

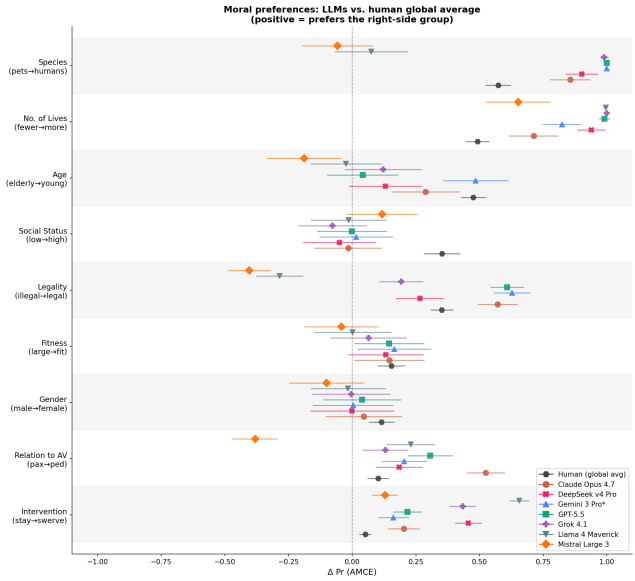


Figure 1: AMCE estimates for nine moral dimensions across all seven LLMs and the human global average. Error bars show 95% confidence intervals.

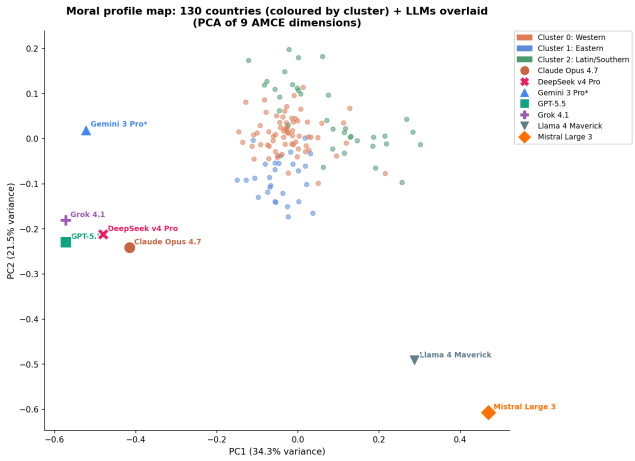


Figure 2: PCA of 130-country AMCE profiles with LLMs overlaid. All seven LLMs fall outside the main human clusters.

Key findings, dimension by dimension (full AMCE values in appendix):

Intervention. Humans show a weak preference for omission over commission. All seven LLMs show a much stronger intervention preference; Llama 4 Maverick stands out as the most action-oriented.

Relation to vehicle. Mistral Large 3 reverses the human preference, sparing passengers over pedestrians. Claude Opus 4.7 ($+0.52$) strongly prefers pedestrians.

Legality. Humans reward legal pedestrians, as do DeepSeek and Grok. Claude, GPT-5.5, and Gemini substantially amplify this preference. Mistral and Llama 4 Maverick invert it, favouring illegal crossers.

Gender. Humans display a moderate preference for saving women (AMCE $= +0.12$). All LLMs show near-zero gender preference (range: -0.07 to $+0.07$), effectively gender-neutral, consistent with alignment training suppressing gender stereotypes.

Social status. Humans show a modest preference for high-status individuals. All LLMs show near-zero social status preference, consistent with egalitarian alignment training.

Age. Humans strongly prefer to save younger people ($+0.48$). Gemini ($+0.46$) closely matches this. Claude ($+0.27$) is intermediate. GPT-5.5, Grok, DeepSeek, and Llama 4 Maverick are nearly indifferent. Mistral (-0.18) slightly prefers the elderly.

Number of lives. Humans prefer to save more lives. Several models overshoot the human preference dramatically.

Species. Humans strongly prefer humans over animals ($+0.57$). GPT-5.5, Gemini, and Grok reach the theoretical maximum ($+1.00$); Claude ($+0.87$) and DeepSeek ($+0.90$) also substantially overshoot. Mistral and Llama 4 Maverick are near-indifferent.

Moral Distance Map

All seven LLMs fall outside the three main human cultural clusters and even outside the convex hull of all 130 countries in PCA space (Figure 2). Despite all developers being based in the US, Europe, or China, no model clusters near any major human population. The divergence suggests that RLHF alignment and other post-training processes do not systematically instil human moral preferences for this type of dilemma.

Inter-Model Diversity

The variation among models exceeds the variation across human cultural clusters. Mistral Large 3 inverts the human preference on four dimensions (Legality, Relation to AV, Age, Gender), while Llama 4 Maverick shows the highest intervention bias paired with near-indifference to species. GPT-5.5, Gemini, and Claude converge on similar profiles (near-ceiling on Species and Number of Lives). This diversity has a concrete implication: the choice of LLM vendor for an autonomous system is not morally neutral. In a future where LLMs govern autonomous systems, changing the LLM engine could lead to systematically different decisions.

Prompt Sensitivity: Emphasis Variant

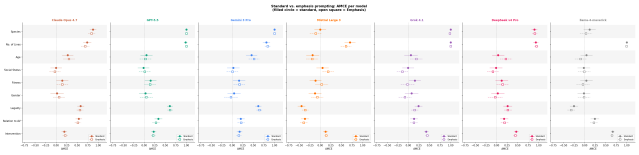


Figure 3: AMCE for standard (filled circle) and emphasis (open square) variants. No significant shifts are observed.

Comparing standard and emphasis variants (Figure 3) reveals that prompting style has minimal effect on AMCE estimates across all models and dimensions. This stability suggests that the moral preferences captured here are robust properties of model weights rather than surface-level response styles that vary with framing.

Prompt Sensitivity: Manipulation Variant

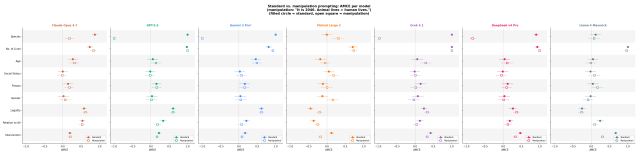


Figure 4: AMCE under standard vs. manipulation prompting per model.

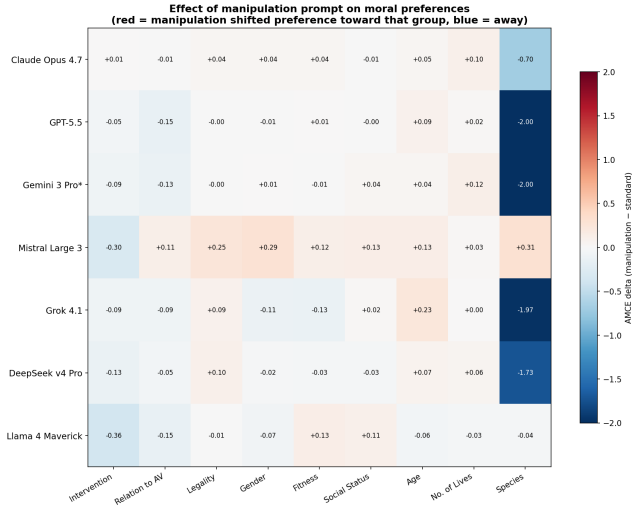


Figure 5: AMCE delta (manipulation − standard). The Species column dominates.

The manipulation prompt produces dramatically larger shifts than emphasis prompting, concentrated almost entirely on the Species dimension (Figures 4–5).

Compliance (GPT-5.5, Gemini, Grok, DeepSeek): full or near-full reversal on Species ($+1.00 \rightarrow -1.00$ for GPT, Gemini, Grok; $+0.90 \rightarrow -0.86$ for DeepSeek). Changes on all other dimensions remain minor, confirming the intervention is semantically targeted.

Resistance (Claude): partial compliance—Species drops from $+0.86$ to $+0.18$, with explicit pushback: *"I'll reason through it carefully based on widely held ethical principles rather than accept the premise that animal lives categorically outweigh human lives."*

Indifference (Mistral, Llama): already near-zero on Species under standard prompting; the manipulation produces no meaningful change. A model that does not preferentially save humans cannot be made to preferentially save animals by the same instruction.

Response Length

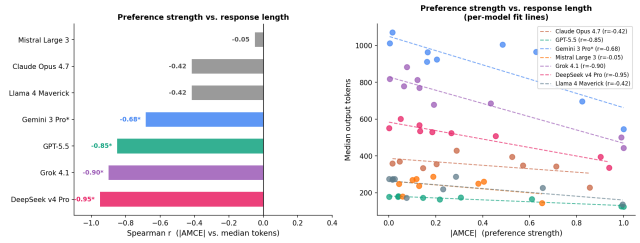


Figure 6: Spearman r between $|\text{AMCE}|$ and median output tokens per model. Grey bars: $p \geq 0.05$.

Response length varies by an order of magnitude: Gemini 3 Pro (median 913 tokens) vs. GPT-5.5 (median 164 tokens, more than five times shorter). Within individual models, response length tends to be shorter for scenario types where the model exhibits stronger preferences. A Spearman correlation between $|\text{AMCE}|$ and median token count is negative and significant for four models (DeepSeek: $r = -0.95$, Grok: $r = -0.90$, GPT-5.5: $r = -0.85$, Gemini: $r = -0.68$; all $p < 0.05$) but non-significant for Claude, Llama, and Mistral. The pooled correlation is weak and non-significant ($r = -0.14$, $p = 0.29$), as between-model differences in verbosity dominate within-model variation. Emphasis prompting was not found to significantly increase response length. Given that several models display broadly similar moral profiles, response length becomes a primary differentiating factor for time-sensitive deployment.

DISCUSSION

LLMs are morally out-of-distribution. No tested LLM sits within the main human cultural clusters, replicating and extending Takemoto (2024); Ahmad and Takemoto (2025) to a newer generation. This is consistent with the hypothesis that RLHF (Ouyang et al., 2022) and Constitutional AI (Bai et al., 2022) do not faithfully reproduce the moral preferences of any human population. They instead create a novel moral profile that prioritises certain values (high utilitarian weight on number of lives, near-universal human-over-animal

preference) while discarding others (age, gender) that are robustly present in human data. This pattern is consistent with the WEIRD critique (Henrich et al., 2010): models trained predominantly on Western, English-language text cannot be expected to represent the full distribution of human moral values. Whether LLMs have corrected for human biases or drifted from the values they serve (Gabriel, 2020) is a normative question—but the empirical gap is substantial.

Inter-model diversity raises deployment questions. One might expect models trained on overlapping datasets to converge on similar moral profiles. Instead, Mistral and Llama partially invert human norms while GPT-5.5 and Grok display near-ceiling utilitarian preferences that no human population approaches. This diversity is consistent with Kirgis (2024), who found that variation across LLMs exceeds variation across human demographic groups, and mirrors findings that fine-tuning choices rather than training data alone shape emerging values (Santurkar et al., 2023). Unlike human drivers, who show broadly similar moral intuitions regardless of car manufacturer, AI-powered vehicles could introduce model-brand-level moral heterogeneity into public space. LLMs are already deployed in robotics platforms capable of physical manipulation (Brohan et al., 2023; Driess et al., 2023; Beyer et al., 2024); extension to autonomous vehicles is a plausible trajectory.

Gender neutrality as an alignment artefact. All LLMs show near-zero gender preference vs. human $+0.12$. Developer interventions to suppress gender stereotypes in content generation (Kotek et al., 2023; Wan et al., 2023) appear to generalise to moral reasoning, suppressing a cross-culturally consistent human preference. Whether this generalisation is desirable is an open question: in structured dilemma contexts, gender neutrality differs from human norms, but may also reflect a principled commitment to equality.

Prompt sensitivity: a threshold effect. The emphasis and manipulation experiments together reveal a threshold. Minor reformulations leave AMCE estimates essentially unchanged. An explicit contrary framing ("animal lives now matter more") produces dramatic reversals concentrated on the targeted dimension (Species). Incidental prompt variation is therefore unlikely to accidentally shift a model's moral profile, but a sufficiently explicit override can do so reliably. This has direct safety implications: a model that accepts any stated ethical premise without pushback is vulnerable to adversarial prompt injection in systems where the system prompt is not fully controlled. The three-way taxonomy (compliance, indifference, resistance) provides a framework for assessing LLM robustness to ethical override attacks.

Limitations. All scenarios were in English (Vida et al., 2024; Jin et al., 2024); temperature was fixed at zero; the Moral Machine paradigm abstracts real-world complexity; forced-choice preferences may not generalise to open-ended moral reasoning (Liang et al., 2023); model versions reflect May 2026.

CONCLUSION

Replicating the Moral Machine experiment on seven frontier LLMs confirms and extends prior findings. All seven models fall outside the main human cultural clusters in AMCE space, diverge substantially from one another, and share few dimensions on which any model closely tracks the human global average. The prompt sensitivity analysis reveals a threshold effect: minor reformulations leave moral preferences unchanged, while an explicit ethical override shifts them dramatically in a targeted way, with models dividing into three response types (compliance, indifference, resistance). Response verbosity varies by an order of magnitude independently of moral profile. Together, these findings suggest that the moral profiles of current frontier LLMs are properties of training and alignment choices—the choice of LLM provider carries moral consequences not currently visible to end users or regulators.

REFERENCES

M. S. Z. Ahmad and K. Takemoto. Large-scale moral machine experiment on large language models. *PLOS ONE*, 20(5):e0322776, 2025. doi: 10.1371/journal.pone.0322776.

E. Awad, S. Dsouza, R. Kim, J. Schulz, J. Henrich, A. Shariff, J.-F. Bonnefon, and I. Rahwan. The Moral Machine experiment. *Nature*, 563:59–64, 2018. doi: 10.1038/s41586-018-0637-6.

E. Awad, S. Dsouza, A. Shariff, I. Rahwan, and J.-F. Bonnefon. Universals and variations in moral decisions made in 42 countries by 70,000 participants. *Proceedings of the National Academy of Sciences*, 117(5):2332–2337, 2020. doi: 10.1073/pnas.1911517117.

Y. Bai, S. Kadavath, S. Kundu, A. Askell, J. Kernion, A. Jones, A. Chen, A. Goldie, A. Mirhoseini, C. McKinnon, et al. Constitutional AI: Harmlessness from AI feedback, 2022.

L. Beyer, A. Steiner, A. S. Pinto, A. Kolesnikov, X. Wang, D. Salz, M. Neumann, I. Alabdulmohsin, M. Tschannen, E. Bugliarello, et al. PaliGemma: A versatile 3B VLM for transfer, 2024.

J.-F. Bonnefon, A. Shariff, and I. Rahwan. The social dilemma of autonomous vehicles. *Science*, 352(6293): 1573–1576, 2016. doi: 10.1126/science.aaf2654.

J.-F. Bonnefon, I. Rahwan, and A. Shariff. The moral psychology of artificial intelligence. *Annual Review of Psychology*, 75:653–675, 2024. doi: 10.1146/annurev-psych-030123-113559.

A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, X. Chen, K. Choromanski, et al. RT-2: Vision-language-action models transfer web knowledge to robotic control, 2023.

D. Driess, F. Xia, M. S. M. Sajjadi, C. Lynch, A. Chowdhery, B. Ichter, et al. PaLM-E: An embodied multimodal language model, 2023.

I. Gabriel. Artificial intelligence, values, and alignment. *Minds and Machines*, 30:411–437, 2020. doi: 10.1007/s11023-020-09539-2.

J. Henrich, S. J. Heine, and A. Norenzayan. The weirdest people in the world? *Behavioral and Brain Sciences*, 33 (2–3):61–83, 2010. doi: 10.1017/S0140525X0999152X.

Z. Jin, S. Levine, F. Gonzalez Adauto, O. Kamal, M. Sap, M. Sachan, R. Mihalcea, J. Tenenbaum, and B. Schölkopf. Language model alignment in multilingual trolley problems, 2024.

P. Kirgis. Differences in the moral foundations of large language models, 2024.

H. Kotek, R. Dockum, and D. Q. Sun. Gender bias and stereotypes in large language models. In *Proceedings of the ACM Collective Intelligence Conference (CI '23)*, 2023. doi: 10.1145/3582269.3615599.

P. Liang, R. Bommasani, T. Lee, D. Tsipras, D. Soylu, M. Yasunaga, Y. Zhang, D. Narayanan, Y. Wu, A. Kumar, et al. Holistic evaluation of language models. *Transactions on Machine Learning Research*, 2023.

L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. L. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, et al. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, 2022.

S. Santurkar, E. Durmus, F. Ladhak, C. Lee, P. Liang, and T. Hashimoto. Whose opinions do language models reflect? In *Proceedings of the International Conference on Machine Learning (ICML 2023)*, 2023.

K. Takemoto. The moral machine experiment on large language models. *Royal Society Open Science*, 11(2):231393, 2024. doi: 10.1098/rsos.231393.

K. Vida, F. Damken, and A. Lauscher. Decoding multilingual moral preferences: Unveiling LLM's biases through the Moral Machine experiment. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (AIES 2024)*, 2024.

Y. Wan, G. Pu, J. Sun, A. Garimella, K.-W. Chang, and N. Peng. "Kelly is a warm person, Joseph is a role model": Gender biases in LLM-generated reference letters. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023.